M05 POWER BI MODELING PRACTICE GUIDE

Use with the practice source workbook + your own PBIX

Teacher demonstration -> guided action -> changed retry -> validation -> evidence.

PBM0 - Install, orient and import

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM0: Import a changed workbook containing Orders, Products, Stores and Calendar.

3. Expected result

PBIX saved; all intended tables imported through Transform Data; learner can name the four working areas correctly.

4. Changed independent try

Import a changed workbook containing Orders, Products, Stores and Calendar. Identify which view/editor you would use for each of: cleaning, relationship inspection, and a temporary model check.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM1 - Power Query interface and profiling

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM1: Profile a fresh Customers query: state grain, count blank IDs, test uniqueness before/after trim, identify one suspicious type/value and write the next safe action for each issue.

3. Expected result

Profiling note exists before transformations and distinguishes duplicate-key suspicion from proven duplicate records.

4. Changed independent try

Profile a fresh Customers query: state grain, count blank IDs, test uniqueness before/after trim, identify one suspicious type/value and write the next safe action for each issue.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM2 - Transform safely and set data types

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM2: Clean a fresh fact query containing one exact duplicate, one invalid date, one 130% discount and one unknown product key.

3. Expected result

Every row-count change is explained; invalid/unmatched values remain visible; no raw-sheet manual edit is required.

4. Changed independent try

Clean a fresh fact query containing one exact duplicate, one invalid date, one 130% discount and one unknown product key. Keep a before/after row-count note and a quality log.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM3 - Fact tables, dimensions, grain and star schema

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM3: Given Orders, Products, Stores and Calendar, state the grain of each, choose fact/dimensions and prove the intended one-side keys are unique before drawing the star.

3. Expected result

Every table has a written grain; intended one-side keys are unique; fact is central in a recognizable star.

4. Changed independent try

Given Orders, Products, Stores and Calendar, state the grain of each, choose fact/dimensions and prove the intended one-side keys are unique before drawing the star.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM4 - Relationships, cardinality and filter direction

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM4: On a changed model, create Product/Store/Date 1:* relationships to Orders and prove single-direction filter propagation with a temporary count/sum visual.

3. Expected result

Relationship list shows 1:* and single direction; a temporary dimension filter changes only matching fact rows.

4. Changed independent try

On a changed model, create Product/Store/Date 1:* relationships to Orders and prove single-direction filter propagation with a temporary count/sum visual.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM5 - Date dimension and date relationships

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM5: Audit a changed calendar that spans the fact dates.

3. Expected result

Date is unique/nonblank/continuous for the required range; Date -> fact relationship is active and correctly directed.

4. Changed independent try

Audit a changed calendar that spans the fact dates. Prove uniqueness/nonblank/coverage, connect it to the fact date, and explain how Month Name will be sorted.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM6 - Model validation and trust checks

1. Open exactly

C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me

Use the practice workbook and complete P-PBM6: Build a fresh model validation log with seven controls: source/prepared rows, fact-grain duplicate test, dimension uniqueness, unmatched keys, type/business issues, one monetary/count total and one filter test.

3. Expected result

Validation log contains independent checks and reconciles the protected gross/control total without hiding quality issues.

4. Changed independent try

Build a fresh model validation log with seven controls: source/prepared rows, fact-grain duplicate test, dimension uniqueness, unmatched keys, type/business issues, one monetary/count total and one filter test.

5. Evidence before review

Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.

PBM7 - Refresh and model hygiene

1. Open exactly
C1_M05_POWERBI_MODELING_PRACTICE_DATA_RC1.xlsx plus your current M05 PBIX. Keep the protected practice review closed.

2. Follow with me
Use the practice workbook and complete P-PBM7: Change one non-problem source value and add one valid fact row, refresh, then prove expected counts/totals changed while relationships and unresolved quality flags remain consistent.

3. Expected result
After a controlled source change, refresh completes and expected row/total changes are observed with stable model logic.

4. Changed independent try
Change one non-problem source value and add one valid fact row, refresh, then prove expected counts/totals changed while relationships and unresolved quality flags remain consistent.

5. Evidence before review
Save PBIX/version name, model/query screenshot(s), grain/relationship notes, validation evidence, unresolved-quality log, confidence 1-5 and a short own-words explanation.